



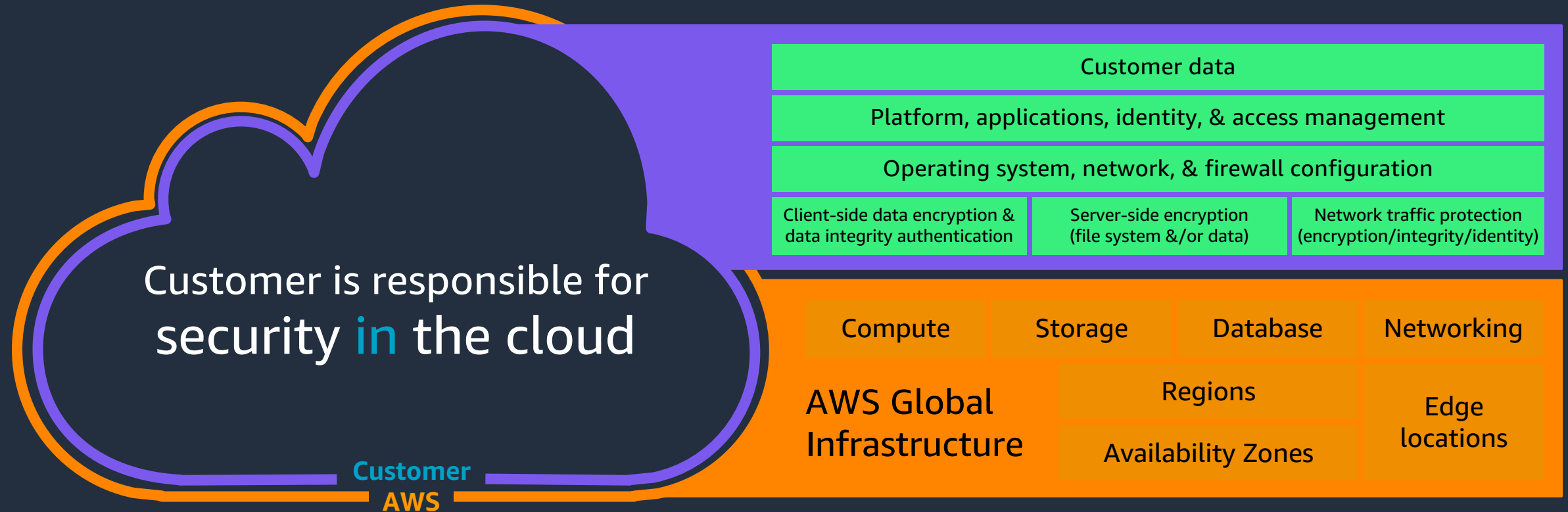
High performance computing on AWS

Solve complex problems with
pay-as-you-go infrastructure

James Grant (he/him/his)

Research Solutions Architect
Amazon Web Services

Share your **security responsibility** with AWS



THE Paris... CLIMATE 10 years PLEDGE Early

Commitment to a sustainable future

On September 19, 2019, Amazon and Global Optimism announced [The Climate Pledge](#), a commitment to meet the Paris Agreement 10 years early. The commitment aims include:

- Net zero carbon by 2040
- Path to 100% renewable energy by 2025
- \$2 billion Climate Pledge Fund
- [Water positive by 2030](#)



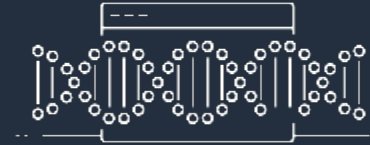
HPC is all around us



Build designs faster
with computational fluid
dynamics (CFD) simulations



Fast-track drug
discovery and structure-
based drug design



Advance genomics insights
using predictive, real-time, or
retrospective data applications



Run geoscientific simulations
and seismic processing
and iterate models faster

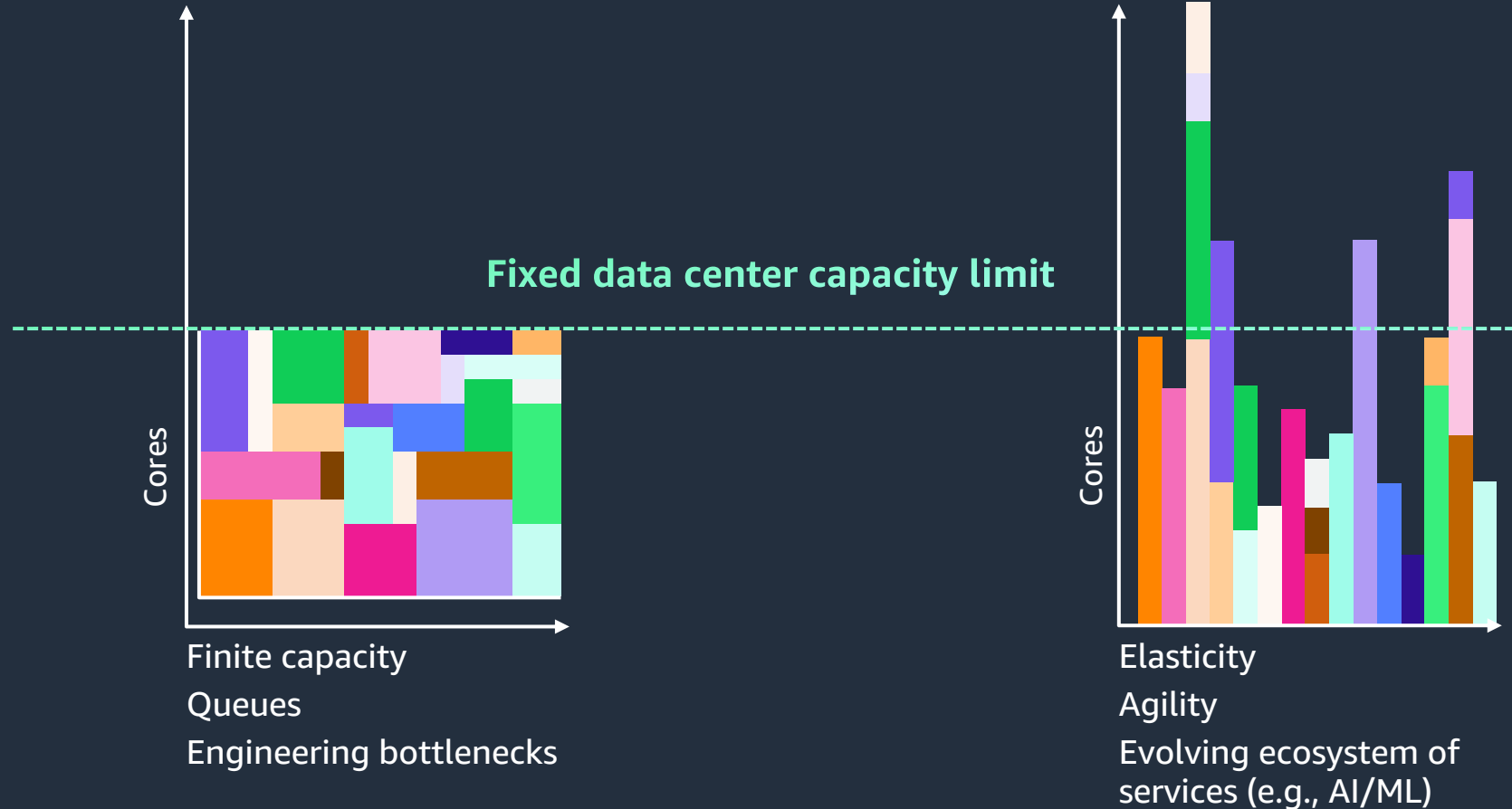


Conduct grid-computing
simulations and identify portfolio
risks and hedging opportunities



Process complex workloads
and analyze massive data
pipelines to further research

What if you could escape the bounds of on-premises?



*Source: Hyperion ROI Study (<http://www.hyperionresearch.com/roi-with-hpc/>)



Amazon EC2 | The compute platform for every workload

Workload types



Machine Learning



High-Performance Computing



Media Rendering



Containers



Web-based Apps



Batch Processing



Big Data

Instance types for HPC workloads

HPC Optimized

Hpc7g

Hpc7a

Hpc6id

Compute, Memory, and Networking

C7gn

C7g

C6i

M6i

R6i

X2iezn

C5n

M5zn

Accelerators

P5

G5

DL1

Trn1

F1

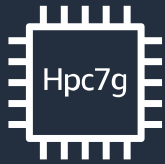
Inf2

VT1

Scale tightly and loosely-coupled HPC applications

- Choice of processor (e.g., Graviton, Intel, AMD)
- Scale tightly-coupled HPC and ML workloads
- Up to 400 Gbps network bandwidth
- < 15 micro-seconds network latencies
- Accelerators use hardware to perform functions more efficiently than is possible in software running in CPUs

HPC-optimized Amazon EC2 instances

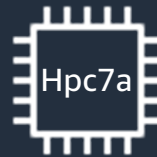


Powered by the next-generation
AWS Nitro System

Amazon EC2 Hpc7g instances

Based on custom AWS Graviton3E processors with low latency and high network performance for MPI-based applications

2.6 GHz 64 cores Graviton3E
128 GB memory
200 Gbps EFA

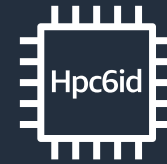


Powered by 4th Gen
AMD EPYC processors

Amazon EC2 Hpc7a instances

Compute-intensive applications like computational fluid dynamics and numerical weather prediction

3.7 GHz and up to 192 cores AMD Genoa
768 GB memory
300 Gbps EFA



Powered by 3rd Gen
Intel Xeon Scalable processors

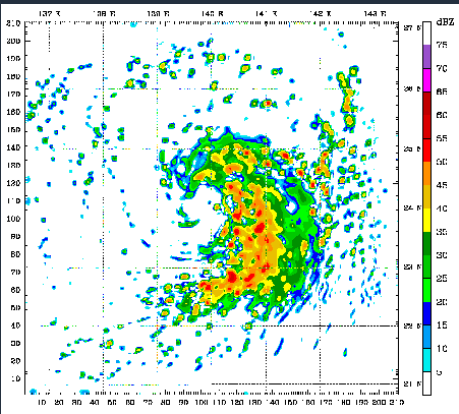
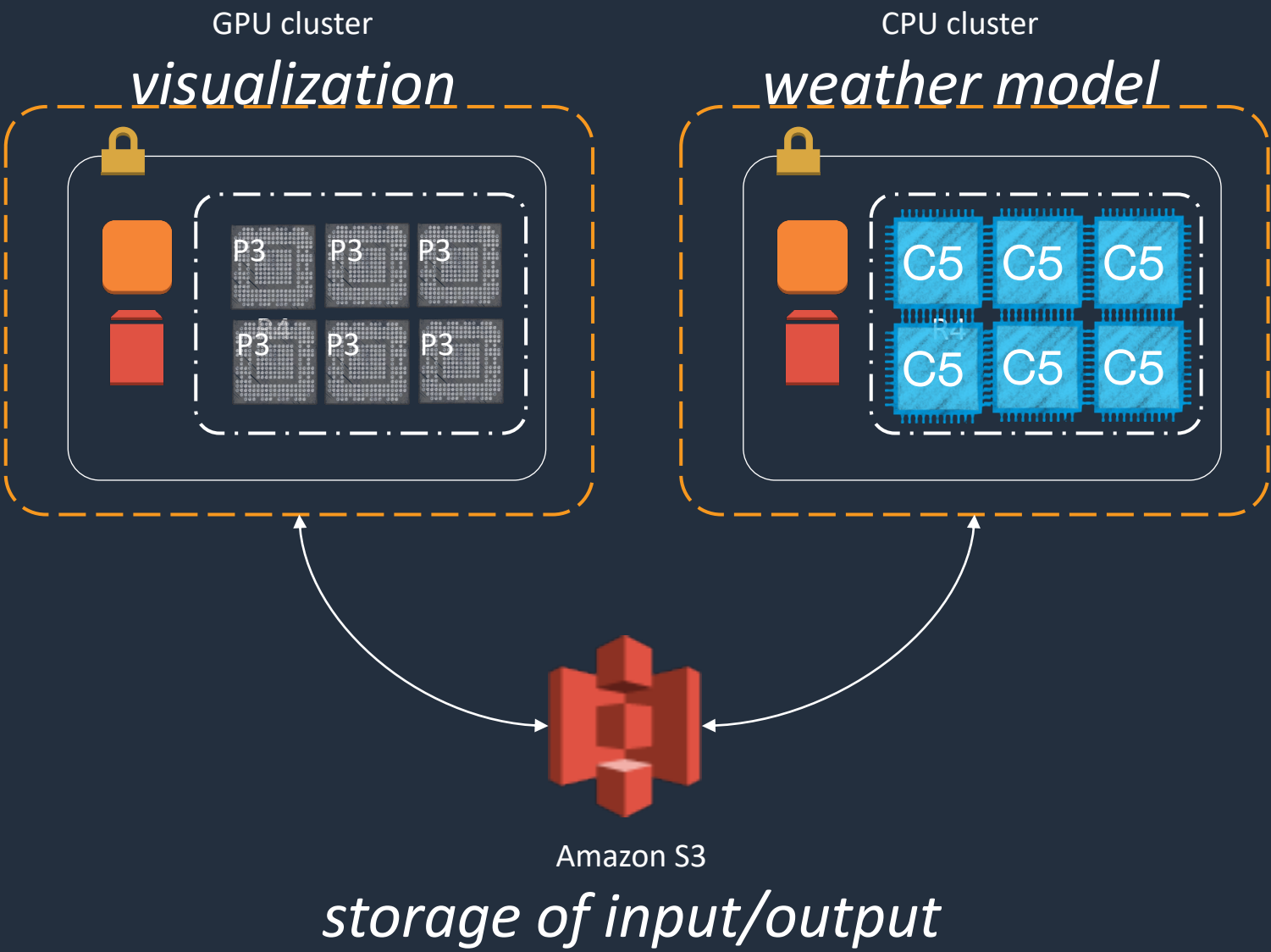
Amazon EC2 Hpc6id instances

Memory-bound and data-intensive workloads like finite element analysis and seismic simulations

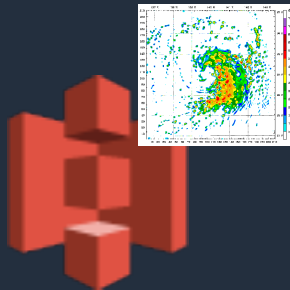
3.5 GHz 64 cores Intel Ice Lake
1,024 GB memory | 15.2 TB NVMe
200 Gbps EFA



Compute clusters in the cloud fit the purpose



Compute clusters in the cloud are ephemeral



Amazon S3

storage of input/output



Elastic Fabric Adapter (EFA)

SRD protocol



Proving myths about latency constraints wrong



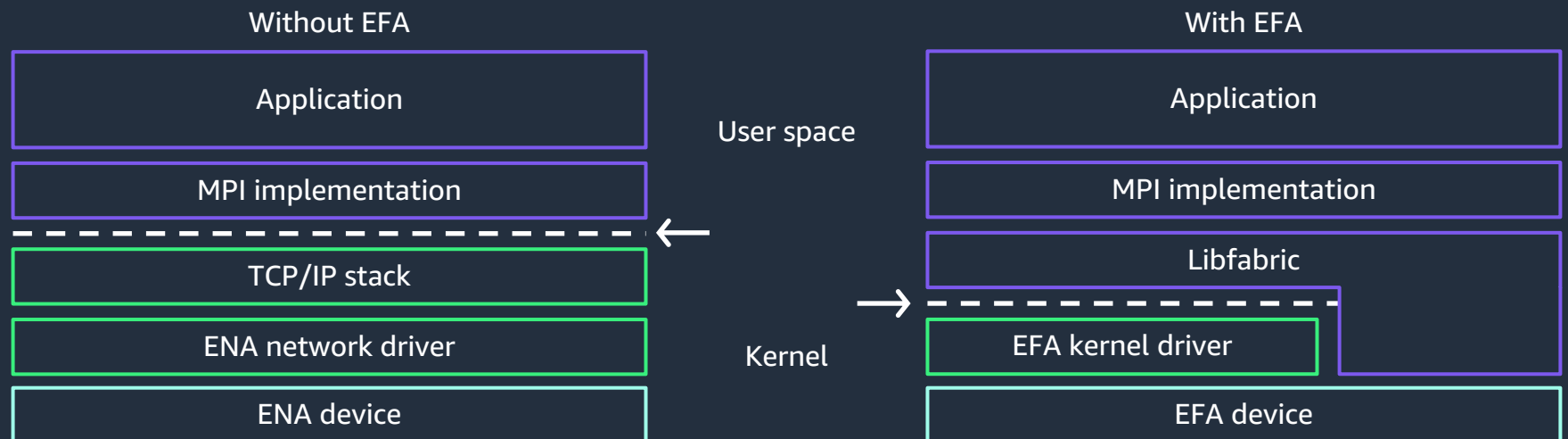
CFD



Seismic



Weather modeling

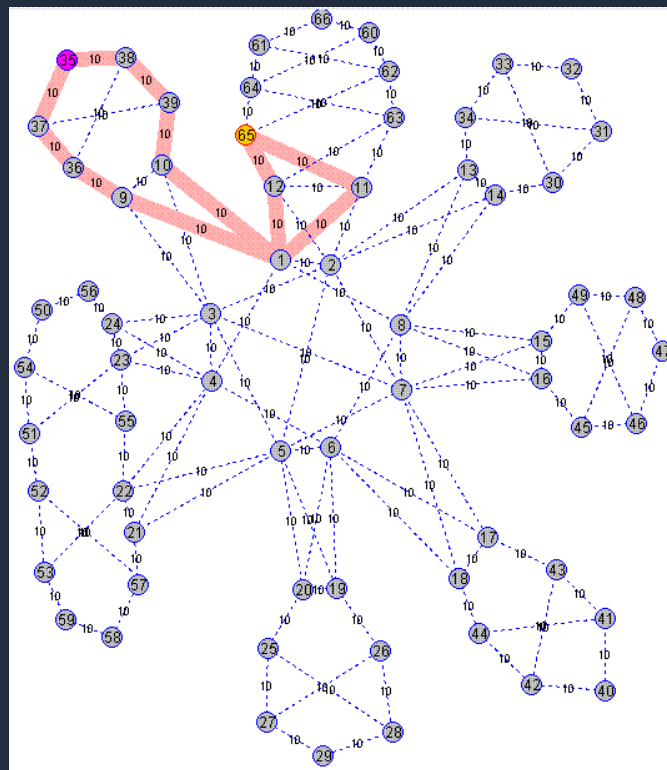


Elastic Fabric Adapter—networks built to scale



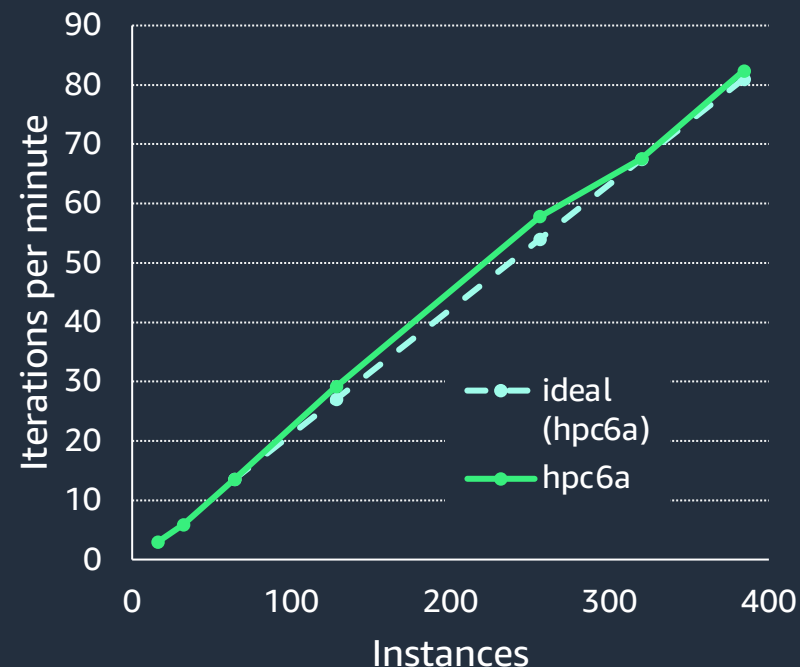
Up to
400 Gbps
networking
bandwidth

- ✓ OS bypass
- ✓ GPUDirect and RDMA
- ✓ Libfabric core supports wide array of MPIs and NCCL



ECMP-enabled packet spraying
and cloud-scale congestion control

1.4B cell Siemens Simcenter STAR-CCM+ automotive CFD simulation



At ~40,000 cores (400 nodes), Hpc6a+EFA shows nearly 100% scaling efficiency

AWS ParallelCluster

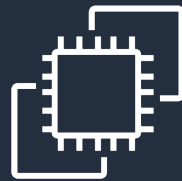
One-stop shop to set up your HPC cluster



Integrated with AWS services you need

FSx

Highly-performant
file systems



Amazon EC2
instances



EFA



NICE DCV

Amazon FSx for Lustre

FULLY MANAGED SHARED STORAGE BUILT ON THE WORLD'S
MOST POPULAR HIGH-PERFORMANCE FILE SYSTEM



Sub-ms latencies, **hundreds of GB/s of throughput**, millions of IOPS



Cost-optimized file systems with
HDD and SSD storage options



Concurrent access for thousands of
instances and **100,000s of cores**



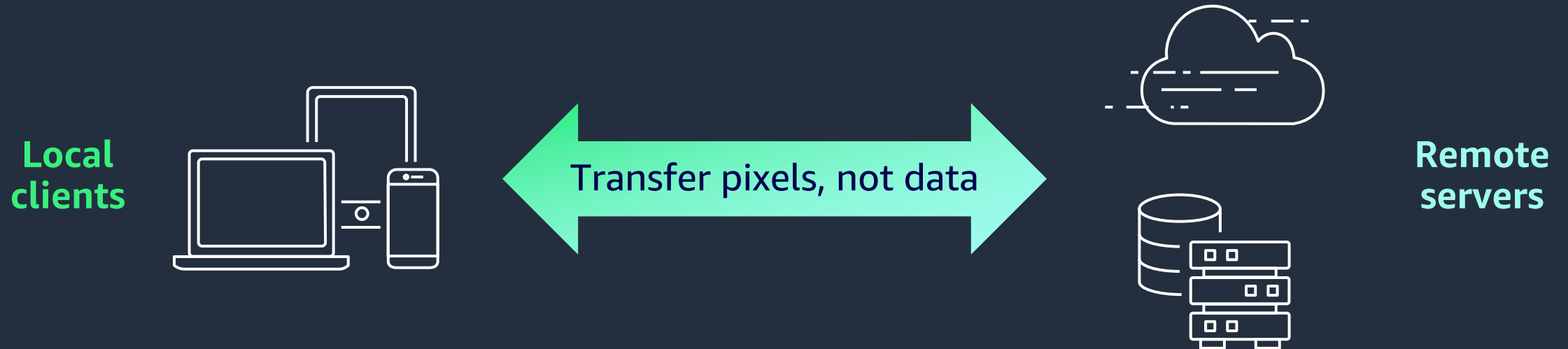
Flexible deployment options for
short- and longer-term workloads

Learn more: Amazon FSx for Lustre, <https://aws.amazon.com/fsx/lustre/>



Amazon DCV

Encrypt and transport pixels to devices



Access, manipulate, and share business-critical information,
regardless of location, over LAN, or WAN networks

Formula 1 redesigns car for closer racing and more exciting fan experience

Formula1 runs its Computational Fluid Dynamics (CFD) platform on AWS HPC

- Reduces downforce loss in wheel-to-wheel racing from 50 percent to 15 percent reducing the impact of the car in front of it
- Better supports strategic priorities for increasing competitiveness and unpredictability on the track
- Cars can drive closer to one another and overtake more easily creating a world-class spectacle for fans

Lowered the cost of CFD simulations by 30%

Reduced CFD simulation time by 80%



CUSTOMER PROFILE



Research on AWS

Compute Scaling Case Studies

[Clemson University](#)

[University of Manchester](#)

[OMass Therapeutics with Yellow Dog](#)

HPC and Research Compute on AWS

[AWS ParallelCluster](#)

[HPCTechShorts](#)

[Ronin](#)

Programs and Credits for Research

[Open Data on AWS](#)

[Global Data Egress Waiver Program](#)

[AWS Cloud Credits for Research](#)



AWS can enable secure research and collaboration, increasing productivity and reducing time to science.